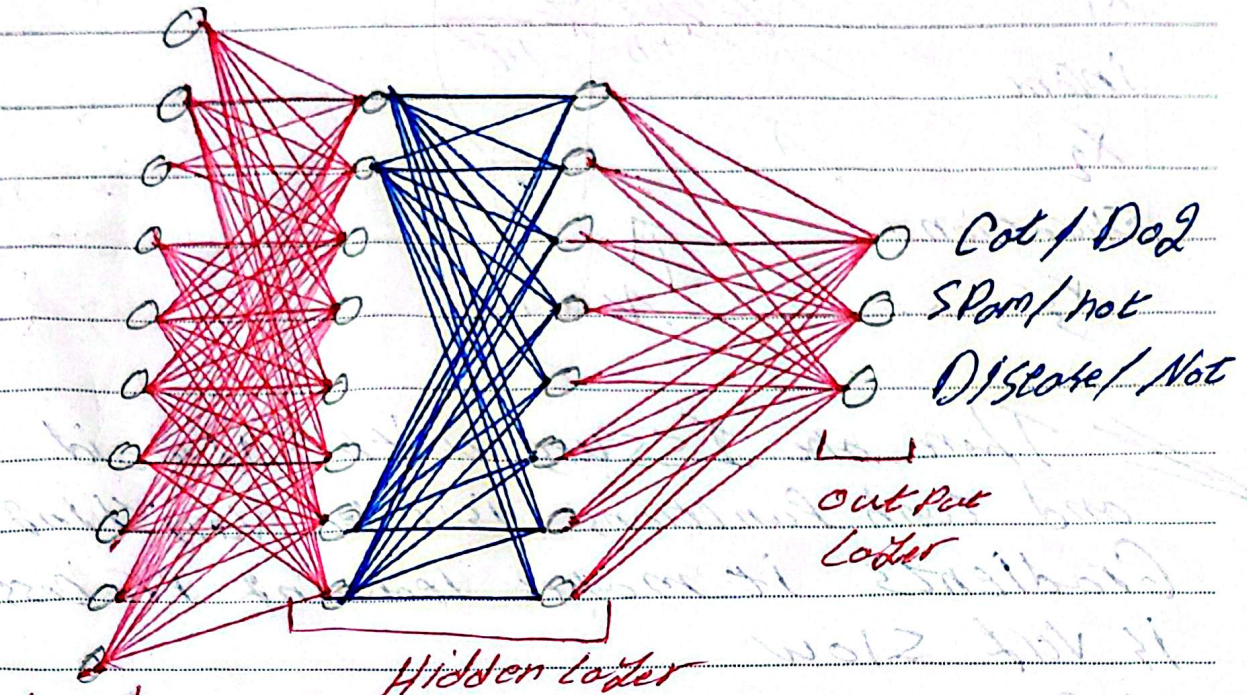


Neural Network

It's a ~~neural~~ machine learning model inspired by the human brain.



Input Layer

Receive data

Age

Height

Weight

or

Images Pixels

Hidden Layer

Process the information

and learn

Patterns

A network

1 - One hidden layer

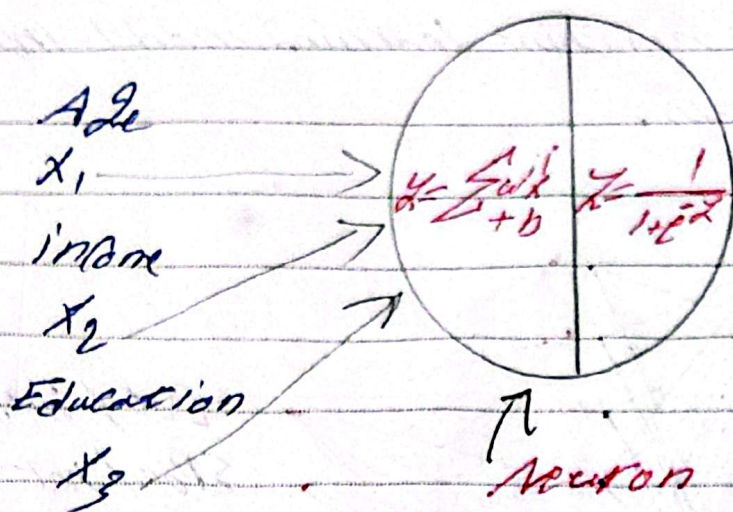
2 - Multiple hidden layer (Deep Learning)

Output Layer

Cat / Dog

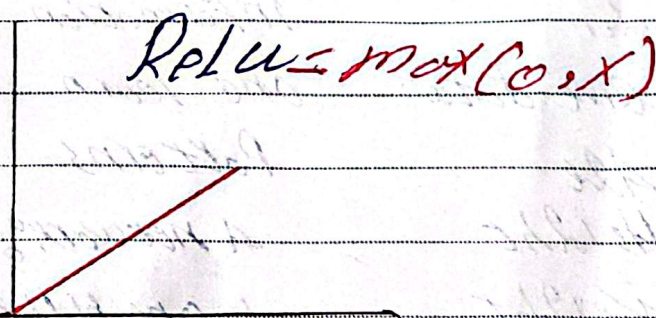
Spam / Not

Disease / Not



There are Issues with Sigmoid and tanh functions. It is called Vanishing Gradients it makes learning process is very slow

For last reason we use $ReLU(x)$



→ For hidden layers if you are not sure which activation fun to use, just use ReLU as your default choice

KAYAN

of Effect of Loss)

$$MAE = \sum_{i=1}^n abs(y_i - \hat{y}_i)$$

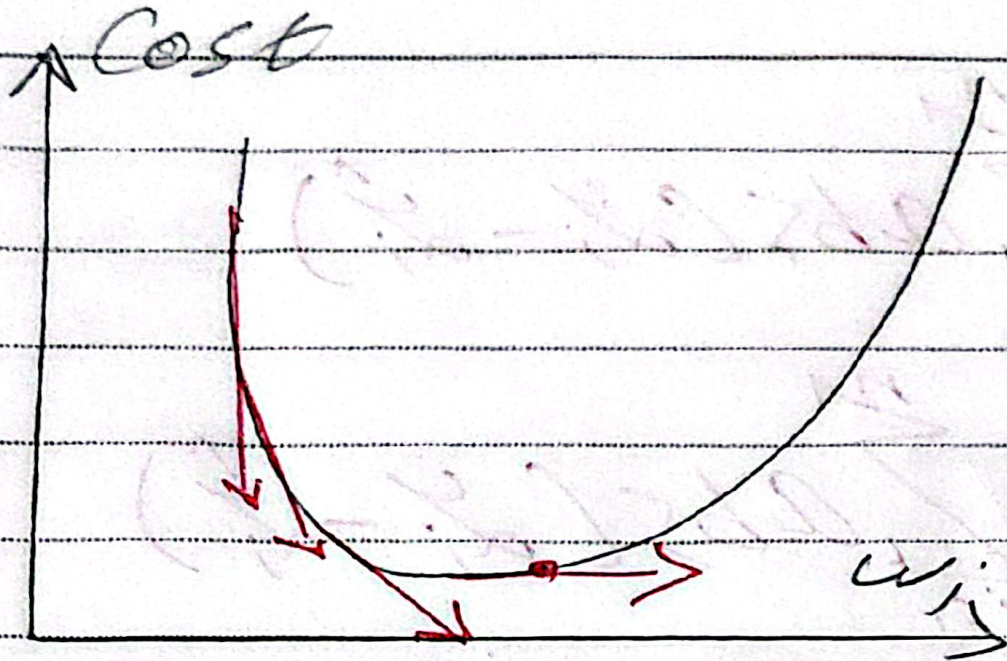
$$MAE = \frac{1}{n} \sum_{i=1}^n abs(y_i - \hat{y}_i)$$

Best Friend

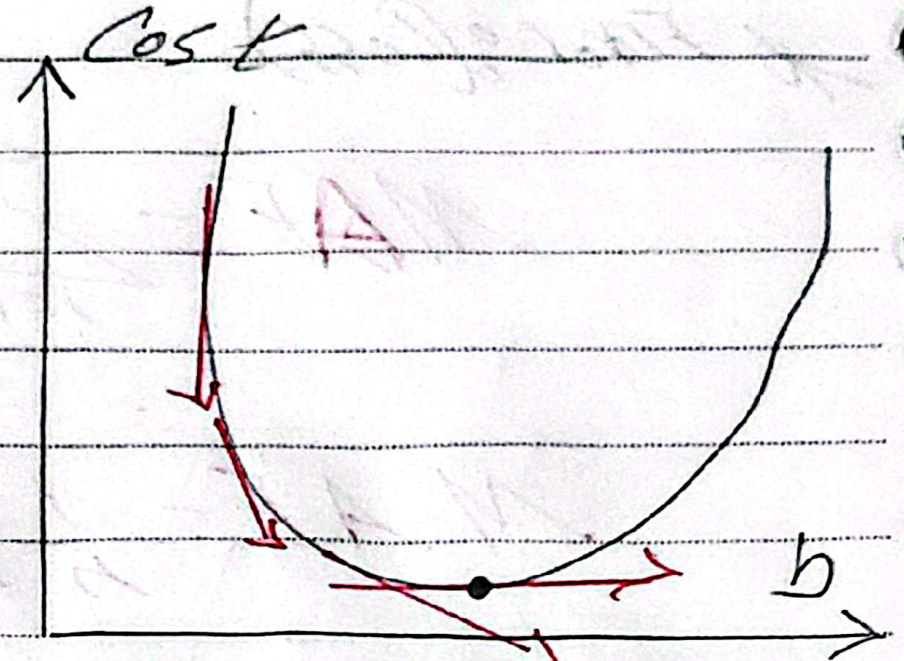
$$MSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

التاريخ: / /

الموضوع:



$$w_j = w_j - \text{LearningRate} \times \frac{\partial}{\partial w_j}$$



$$b = b - \text{LearningRate} \times \frac{\partial}{\partial b}$$

